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ABSTRACT

Recent advances in X-ray free-electron laser diagnostics have enabled direct probing of electronic properties under extreme pressures and temperatures, such as those encountered in stellar interiors and inertial confinement fusion experiments, challenging theoretical models for interpreting experimental data. Kohn–Sham density functional theory (KSDFT) has been successfully applied to analyze experimental X-ray scattering measurements, but its high computational cost renders routine application impractical. Orbital-free DFT (OFDFT) is a substantially more efficient alternative, with computational cost scaling linearly with system size and a weak temperature dependence, yet it often lacks the accuracy required for describing the electronic density and the electron–ion structure factor. Overcoming this limitation, we present a non-empirical KS-assisted orbital-free density functional framework for calculations under extreme conditions, which enables efficient OFDFT simulations with KSDFT-level accuracy for electron densities, electron–ion structure factors, and equations of state across a broad range of conditions. Benchmark comparisons with quantum Monte Carlo data for dense hydrogen and validation against Rayleigh weight measurements of hot dense beryllium demonstrate the reliability of the framework and speedups of up to several hundred times compared with KSDFT. We further show that even at temperatures of the order of 100 eV, quantum nonlocality remains essential for correctly describing the electron–ion structure factor in dense hydrogen.

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I. INTRODUCTION

Matter exposed to extreme temperatures and pressures—reaching thousands to millions of kelvin and densities comparable to or exceeding those of solids—plays a central role in the interiors of planets and stars. These conditions can now be explored in the laboratory using large-scale facilities that combine high-power lasers with advanced spectroscopic diagnostics, such as the European XFEL,¹ SACLA,² SLAC,³ and the National Ignition Facility

(NIF).⁴ Together, these platforms enable direct experimental access to matter under astrophysically relevant conditions.

Understanding material behavior in this regime is essential for modeling giant planets,⁵ white dwarfs,⁶ and other stellar objects.⁷ Extreme states of matter are also of key importance for inertial confinement fusion (ICF) applications, and open opportunities to discover materials with unusual or previously inaccessible properties.^{8–10} Recent milestones include the first experimental creation of liquid carbon,¹¹ the observation of superheating

in gold crystals beyond the entropy-catastrophe limit,¹² laser-induced phonon hardening in gold,¹⁰ and the production of warm dense beryllium exhibiting pressure-driven K-shell delocalization.⁴ Progress in this field is particularly motivated by inertial fusion energy, highlighted by the recent demonstration at the NIF of fusion energy output exceeding the input laser energy.¹³

Among the extreme states of matter, the warm dense matter (WDM) regime is particularly challenging. In WDM, thermal excitations, quantum effects, and strong interparticle interactions are all comparable in importance, leading to a complex interplay that governs material properties. This mixed character strains the assumptions of standard theoretical models and often pushes established approaches beyond their limits.^{7,14–16} These challenges have become particularly acute with the rapid development of advanced diagnostic techniques that employ various types of X-ray probes to measure WDM properties. In particular, modern X-ray free-electron laser diagnostics have advanced to the point where they can directly probe electronic properties at various length scales, rather than being limited to macroscopic quantities such as the equation of state (EOS). One example is the Rayleigh weight,¹⁷ which describes electronic localization around the nuclei and is closely related to the electron–ion static structure factor.

Kohn–Sham (KS) density functional theory (KSDFT)^{18,19} remains the primary *ab initio* framework for modeling electronic properties in the WDM regime. However, its computational cost increases rapidly with temperature, since an increasingly large number of KS orbitals is required to accurately capture thermal electronic excitations. As a result, it is not uncommon for the analysis of datasets from a single experimental campaign using KSDFT to require years of combined computational and human effort, including parameter optimization, repeated simulations, and data analysis at each stage. This creates a significant bottleneck and often forces researchers to rely on lower-level, semi-analytic models for data interpretation.

Orbital-free density functional theory (OFDFT)^{20–23} is an alternative, computationally much cheaper approach that eliminates the need to explicitly compute orbitals. As a result, the computation time of OFDFT calculations scales linearly with system size and has negligible temperature dependence,^{24,25} making it well suited for studying WDM and dense plasmas over a wide region of phase space. The primary factor determining the accuracy of OFDFT calculations is the approximation used for the non-interacting free-energy functional, which is calculated exactly within KSDFT. Early OFDFT simulations of WDM and dense plasmas relied on simple, low-level approximations, such as the Thomas–Fermi model (TF) with gradient corrections,^{26,27} which are accurate only in the extreme limits of very high temperature and density. More recent free-energy functionals are typically constructed to reproduce ion–ion pair correlation functions or macroscopic observables such as the EOS.^{21,28–31} However, agreement with these metrics does not provide validation of accuracy at the electronic-density level. Therefore, the X-ray diagnostic-driven demand for accurate electron–ion structure factor modeling under extreme pressure and temperature conditions calls for a re-evaluation of the standard methodologies used to test and develop non-interacting free-energy functionals in OFDFT.

To overcome these limitations, we introduce an OFDFT approach that significantly enhances the accuracy of modeling the electron density and the electron–ion structure factor in WDM

and dense plasmas. This non-empirical method optimizes the non-interacting free-energy functional to achieve KSDFT-level accuracy in electron density, all the while maintaining the computational advantages of OFDFT. Our framework consistently delivers accurate electron density, electron–ion structure factor, and reliable EOS. This capability is made possible by a density-driven inversion scheme guided by a low-cost KSDFT calculation carried out on a minimal reference system. We refer to this method as the scalable KS-assisted non-interacting functional for electronic density under extreme conditions, or SKANEX. The SKANEX method is highly efficient and can be applied across a wide variety of materials and thermodynamic conditions. By rigorously benchmarking against both KSDFT and path-integral Monte Carlo (PIMC) calculations^{32–34} over a wide range of temperatures and densities, we demonstrate the high accuracy of SKANEX for warm dense hydrogen. Furthermore, we demonstrate the application of SKANEX by analyzing recent experimental Rayleigh weight data for hot dense beryllium measured at the NIF.^{4,35}

II. THEORY AND METHOD

A. Finite-temperature orbital-free density functional theory

KSDFT provides the standard first-principles framework for electronic-structure calculations in WDM. Both KSDFT and OFDFT are rooted in the Hohenberg–Kohn theorems and their finite-temperature generalization by Mermin.³⁶ In KSDFT, the interacting system is mapped onto a non-interacting reference system with the same electron density, and the non-interacting free energy is evaluated from KS orbitals and Fermi–Dirac occupations. This orbital representation is accurate but becomes increasingly expensive at elevated temperatures, because the number of partially occupied orbitals required for convergence grows rapidly with temperature, approximately as $O(T^{3/2})$. As a result, the cost of KSDFT becomes a major bottleneck for systematic simulations over broad WDM phase spaces. OFDFT circumvents this bottleneck by replacing the orbital expression for the non-interacting free energy with an explicit non-interacting free-energy density functional of the electron density, thereby enabling simulations with near-linear scaling in system size and only weak temperature dependence.

In KSDFT and OFDFT, the free-energy decomposition reads

$$F[n; T] = F_s[n; T] + F_H[n] + F_{XC}[n; T] + F_{ie}[n], \quad (1)$$

where $F_s[n; T]$, $F_H[n]$, $F_{XC}[n; T]$, and $F_{ie}[n]$ denote the non-interacting free energy, Hartree energy, exchange–correlation (XC) free energy, and ion–electron interaction energy, respectively. The central difference between KSDFT and OFDFT is the treatment of F_s : KSDFT evaluates it from orbitals (exactly for a given KS system with a given XC functional), whereas OFDFT approximates it as an explicit functional of the electron density. The Hartree term has the same functional form in both approaches. The XC and ion–electron terms can also be chosen consistently in both methods, although OFDFT is restricted to orbital-independent XC approximations and usually employs local pseudopotentials, because orbitals are not available. Regarding the XC term, standard calculations frequently employ zero-temperature (temperature-independent) XC approximations evaluated at the finite-temperature density. The

temperature-dependent XC functionals have been developed for WDM,³⁷ and recent studies indicate that the differences between finite-temperature and zero-temperature XC corrections typically modify thermodynamic properties by only a few percent, as observed in warm dense beryllium³⁸ and warm dense hydrogen at the parameter values considered here.^{33,34} In practical benchmarks, using the same XC functional and pseudopotential in KSDFT and OFDFT allows us to isolate the effect of $F_s[n; T]$.

The finite-temperature non-interacting free-energy functional $F_s[n; T]$ is known exactly only through the orbital representation of the non-interacting KS reference system. Developing accurate explicit approximations for $F_s[n; T]$ remains the central methodological challenge in OFDFT. Progress in constructing finite-temperature $F_s[n; T]$ functionals has largely built upon concepts originally developed for ground-state kinetic-energy density functionals, including exact constraints, gradient expansions, and the response properties of the uniform electron gas.^{21,22} Existing $F_s[n; T]$ can be broadly grouped into two classes: semilocal functionals, which use local density variables and their gradients, and non-local functionals, which incorporate density response information through kernel constructions.

For semilocal $F_s[n; T]$, the orbital dependence is replaced by functions of the local density and a small set of local descriptors, such as the reduced density gradient and reduced density Laplacian. A typical finite-temperature generalized gradient approximation (GGA) form can be written as^{21,22}

$$F_s[n; T] = \int \tau_0^{\text{TF}}(\mathbf{r}) [\xi(t) F_\tau(s_\tau) - \zeta(t) F_\sigma(s_\sigma)] d\mathbf{r}, \quad (2)$$

where τ_0^{TF} is the ground-state TF kinetic-energy density, $t = T/T_F$ is the reduced temperature, and $\xi(t)$ and $\zeta(t)$ are the finite-temperature scaling functions for the kinetic and entropic contributions, respectively. The quantities F_τ and F_σ are the corresponding enhancement factors evaluated at the temperature-dependent reduced density gradients s_τ and s_σ .²³ Laplacian-dependent semilocal functionals extend this construction to the meta-GGA level by including the reduced density Laplacian.³⁹ These functionals are motivated by finite-temperature gradient expansions and by the extension of functional forms originally developed for ground-state kinetic-energy functionals to finite-temperature free-energy functionals.^{28,31,40} They have been successfully used in some WDM applications; however, as semilocal approximations, they are not generally constructed to recover the full finite-temperature Lindhard response beyond the long-wavelength limit. Representative semilocal finite-temperature functionals include VT84F⁴¹ and LKTF.²⁸

For a nonlocal non-interacting free-energy functional, $F_s[n; T]$ is decomposed as

$$F_s[n; T] = F_s^{\text{TF}}[n; T] + F_s^{\text{vW}}[n; T] + F_s^{\text{NL}}[n; T], \quad (3)$$

where $F_s^{\text{TF}}[n; T]$ is the TF functional, which is accurate in the high-density or high-temperature limit,⁴² and $F_s^{\text{vW}}[n; T]$ is the von Weizsäcker functional, which is exact in the low-density limit for a single occupied orbital.⁴³ $F_s^{\text{NL}}[n; T]$ contains the remaining nonlocal contribution. $F_s^{\text{NL}}[n; T]$ is commonly constructed to reproduce the finite-temperature Lindhard response in the uniform-electron-gas limit. This constraint has been shown to provide adequate

static density response properties for metallic systems.^{44,45} The class of nonlocal functionals based on the Lindhard response function includes Wang–Teter-type functionals^{46,47} as well as line-integral-based approaches^{29,48–51} (see Refs. 21 and 22 for a broader overview). The strategy of constructing F_s^{NL} from linear density-response theory can be naturally extended by incorporating constraints derived from quadratic and higher-order response functions in the weak-density-perturbation limit.⁵²

Leveraging these developments, OFDFT has been successfully applied to the calculation of EOS and other thermodynamic properties of WDM.^{23,29} In addition to thermodynamic properties, the present work aims to extend the applicability of OFDFT to accurately compute the electron density and the electron–ion static structure factor. To this end, we build upon the state-of-the-art line-integral-based Xu–Wang–Ma functional (XWMF) $F_s^{\text{NL}}[n; T]$ ²⁹ and introduce a new optimization strategy that utilizes KSDFT electron densities. This choice provides a physically well-founded baseline, so that the KS-assisted correction refines an existing nonlocal functional rather than fully replacing it.

B. SKANEX method

The core idea is to introduce a system-dependent correction to the non-interacting free-energy functional, calibrated against KSDFT reference data, enabling OFDFT to achieve KSDFT-level accuracy for electron density. We use the following form of the nonlocal part of the non-interacting free-energy functional:

$$F_{\text{SKANEX}}^{\text{NL}}[n; T] = F_0^{\text{NL}}[n; T] + \beta F_1^{\text{NL}}[n; T], \quad (4)$$

where $F_0^{\text{NL}}[n; T]$ corresponds to the zeroth-order term of the XWMF functional,²⁹ equivalent to a finite-temperature Wang–Teter functional,^{29,47} utilizing a kernel (the second-order functional derivative⁴⁵) that depends only on the mean valence electron density. The term F_1^{NL} represents a first-order correction to $F_0^{\text{NL}}[n; T]$ arising from density inhomogeneity at the level of the non-interacting free-energy kernel and derived using the line integral scheme^{29,49–51} in the XWMF functional. Equation (4) differs from previous models through the introduction of a constant, system-dependent regularization factor β , which controls the magnitude of the correction $F_1^{\text{NL}}[n; T]$.

The parameter β is determined by a density-based KS-assisted optimization. For a given material and thermodynamic condition, one value of β is optimized by minimizing the integrated absolute deviation between the OFDFT and KSDFT electron densities,

$$\min_{\beta} |\Delta n| = \min_{\beta} \int |n_{\text{OF}}(\mathbf{r}) - n_{\text{KS}}(\mathbf{r})| d\mathbf{r}. \quad (5)$$

For each considered configuration of ions, a KSDFT single-point calculation provides the reference electron density, and the corresponding β_i is obtained by Powell minimization^{53,54} of the density deviation. The final SKANEX parameter is the arithmetic mean, $\beta = N_i^{-1} \sum_i \beta_i$ (where N_i is the number of ion snapshots), and the uncertainty is estimated as the standard deviation across the selected configurations. This restrained one-parameter calibration reduces the risk of overfitting while retaining the physical limits of the underlying nonlocal functional. In the examples considered in this work, $N_i = 10$ ionic configurations are selected from an OFDFT–molecular dynamics (MD) trajectory generated with

XWMF. The corresponding β values and their standard deviations are listed in [Appendix C](#).

C. Numerical details

Calculations were performed using the ATLAS package^{55,56} for OFDFT and Quantum ESPRESSO (v7.2.0)^{57,58} for KSDFT. In the OFDFT and KSDFT calculations, the electronic exchange–correlation interactions were treated using the Perdew–Zunger local density approximation (LDA)⁵⁹ within the adiabatic approximation.

For hydrogen, ion–electron interactions were modeled via Heine–Abarenkov pseudopotentials.^{31,60,61} In OFDFT calculations, grid spacings of 0.10 and 0.05 Å were employed for MD and EOS studies, respectively. For KSDFT–MD simulations, a kinetic-energy cutoff of 200 Ry and the Γ point were used for $r_s = 1$, whereas a 300 Ry cutoff and a $2 \times 2 \times 2$ k -point mesh were adopted for $r_s = 2$ and 3.23. Static EOS calculations employed a 300 Ry cutoff with a k -point spacing of 0.016 Å^{−1}.

For beryllium, a local, s -channel-only norm-conserving pseudopotential⁶² ($r_c = 0.06$ bohr) was generated using fhi98PP.⁶³ A grid spacing of 0.035 Å was adopted for the OFDFT simulations.

MD simulations were performed in the NVT ensemble using a Nosé–Hoover thermostat.^{64,65} OFDFT utilized the internal MD module of ATLAS, while KSDFT runs were driven by the Atomic Simulation Environment (ASE)⁶⁶ via the QEpy interface.⁶⁷ The time steps were set to 0.1 fs for hydrogen and 0.01 fs for beryllium. For specific conditions at $r_s = 1$, the time step for hydrogen was reduced to ensure numerical stability: it was set to 0.02 fs for $\theta = 1$, and further decreased to 0.003 fs for $\theta = 2$ and $\theta = 4$. The PIMC results were obtained by performing simulations with 14 and 32 particles, and some of the PIMC data were adopted from Ref. 32. Accordingly, to enable a comparison under identical simulation parameters, 14 atoms were employed at $r_s = 2$ and 3.23, and 32 atoms at $r_s = 1$. To assess finite-size effects, additional calculations were performed with 108 and 1024 particles. For beryllium, a fixed size of 64 atoms was used. In all simulations, following 10 000 equilibration steps, statistics were collected over at least 40 000 steps.

Details of the computational efficiency of SKANEX are provided in [Appendix A](#).

The computed static structure factors $S_{ab}(q)$ are defined as

$$S_{ab}(\mathbf{q}) = \frac{1}{\sqrt{N_a N_b}} \langle n_a(\mathbf{q}) n_b^*(\mathbf{q}) \rangle, \quad (6)$$

where $n_{a/b}(\mathbf{q})$ denotes the reciprocal-space density of species a/b . The static structure factor calculations were performed on a three-dimensional uniform grid: directly within ATLAS for OFDFT, and via an on-the-fly QEpy interface for KSDFT. The averaged $S_{ab}(q)$ was computed by averaging over wavevectors with equal magnitude ($|\mathbf{q}| = q$) and over the sampled MD steps.

III. RESULTS

Numerous approximations to the non-interacting free-energy functional have been proposed.²¹ When developing these models, it is standard practice to focus primarily on their ability to reproduce real-space ionic structure and EOS properties. However, by analyzing commonly used functionals, we found that this strategy does not

necessarily yield an accurate description of the underlying electronic properties. Notably, for warm dense hydrogen, conventional state-of-the-art finite-temperature non-interacting free-energy functionals cannot accurately describe the electron–ion static structure factor across a wide density regime, from solid to metallic densities, spanning $\rho \approx 0.08$ g/cm³ up to 0.34 g/cm³. By contrast, SKANEX is specifically designed to provide accurate predictions not only for the EOS, but also for electronic properties, as characterized by the electron density distribution and the electron–ion static structure factor. The latter is particularly important, because it directly determines the Rayleigh weight measured in X-ray scattering experiments,^{4,17,35} and thus is relevant for the diagnostics of material properties in WDM experiments.

We first demonstrate the application of SKANEX to warm dense hydrogen, a material of strong scientific interest because of its relevance to ICF experiments and astrophysical applications. For that, we use standard parameters used to characterize the electronic state: the coupling (density) parameter r_s , which is defined as the ratio of the mean interelectronic distance to the Bohr radius, and the degeneracy parameter $\theta = T/T_F$, which represents the ratio of the temperature to the electronic Fermi temperature.

Figure 1(a) depicts the temperature–density range (highlighted in orange and lying beyond the estimated 50% ionization line) in which SKANEX achieves KSDFT-level accuracy for warm dense hydrogen, as assessed by agreement in the electron density distribution, the electron–ion static structure factor, and the EOS. We illustrate this performance for three density–temperature regimes relevant to current experimental efforts: a high-density case with $\rho = 2.7$ g/cm³ ($r_s = 1.0$), a characteristic metallic-density case with $\rho = 0.34$ g/cm³ ($r_s = 2.0$), and a low-density regime with $\rho = 0.08$ g/cm³ ($r_s = 3.23$), which is typical of hydrogen jet experiments.^{68,69} In all cases, the temperatures are chosen such that the degeneracy parameter is $\theta = 1$. Figure 1(b) illustrates the degree of electron localization around protons under these conditions, showing the electron number density computed using KSDFT for a representative snapshot containing 14 ($r_s = 3.23$ and 2) or 64 ($r_s = 1$) protons. Here, the degree of electron localization is characterized by the ratio $n(r)/n_0$, where n_0 denotes the spatially averaged electron number density. A ratio significantly greater than unity in the vicinity of proton sites indicates strong electron localization around individual protons, whereas a nearly uniform distribution corresponds to a more delocalized, plasma-like state.⁷⁰ The degree of localization is also qualitatively correlated with the ionization state of the system. At the highest density ($r_s = 1$), pressure and thermal ionization lead to a strongly delocalized electron distribution, characteristic of the hot dense plasma phase of hydrogen. As the coupling parameter increases to $r_s = 2$ (i.e., as the density decreases), substantial electron localization around the protons emerges, which is typical of the WDM regime. With a further decrease in density along the line $\theta = 1$, corresponding to $r_s = 3.23$, the system enters a partially ionized regime in which most electrons are localized around individual protons, consistent with the formation of atomic-like states.

Figure 1(c) presents the electron–ion static structure factor in warm dense hydrogen for $r_s = 1, 2$, and 3.23 at $\theta = 1$. Figure 1(d) shows the electron density distributions along the [110] direction of the cubic simulation box. We compare the SKANEX results with corresponding PIMC and KSDFT data for the electron–ion static

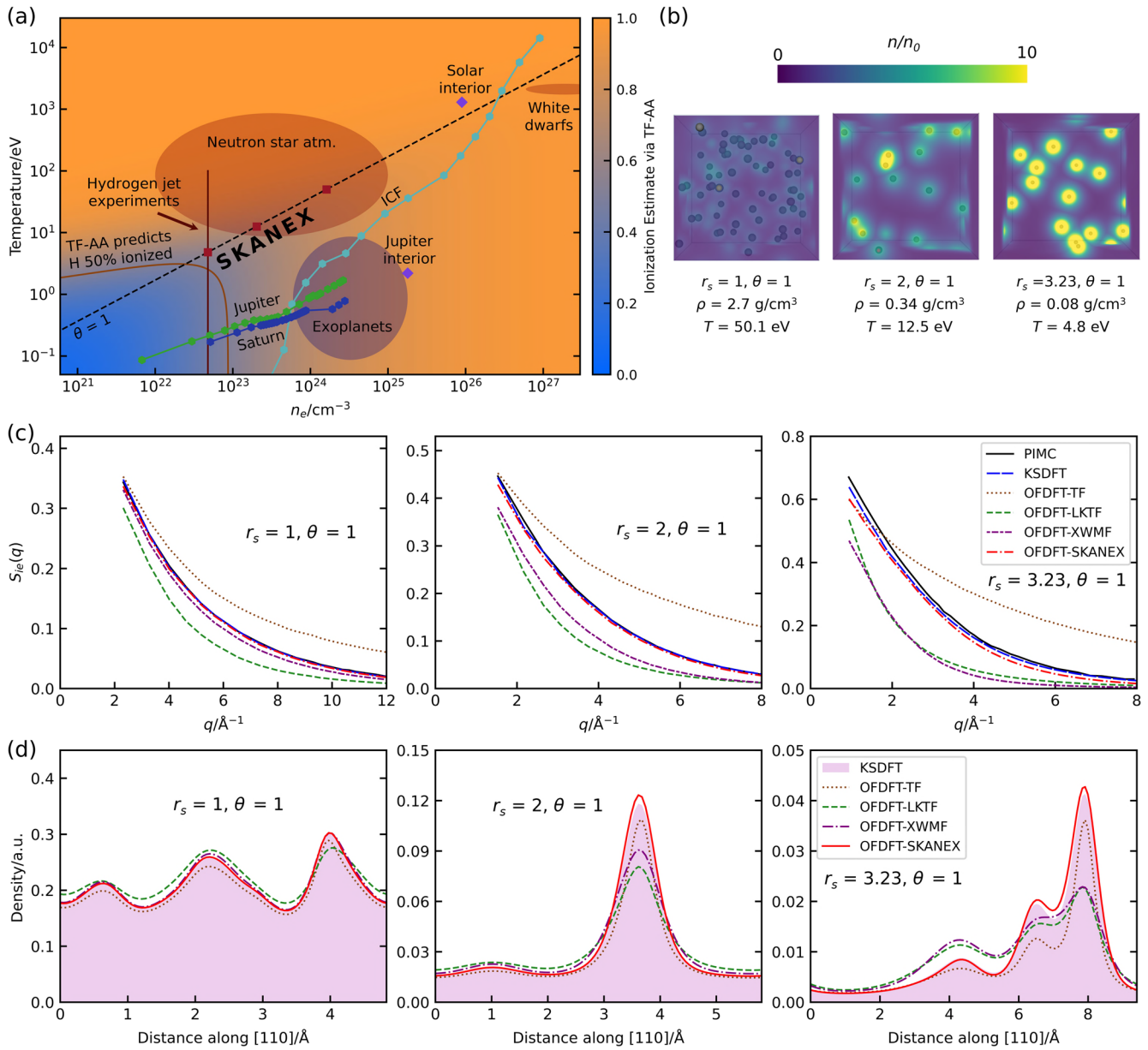


FIG. 1. Results for warm dense hydrogen. (a) Temperature–density plane showing the 50% ionization line evaluated using the Thomas–Fermi average-atom (TFAA) model.⁷¹ Also shown are thermodynamic conditions relevant to neutron-star atmospheres,^{72,73} the interiors of Saturn⁷⁴ and Jupiter,^{75,76} exoplanets,⁷⁷ the solar interior,⁷⁸ white dwarfs,⁷³ inertial confinement fusion (ICF) experiments,⁷⁹ and hydrogen jet experiments.^{68,69} (b) Illustration depicting the changes in the degree of localization of electron density around protons with changes in density, while keeping the electron degeneracy degree constant at $\theta = 1$. The data were generated by performing KSDFT calculations at $r_s = 1, 2$, and 3.23 . (c) Electron–ion static structure factors $S_e(q)$ calculated using PIMC, KSDFT, and OFDFT, for $r_s = 1, 2$, and 3.23 at $\theta = 1$. PIMC results for $r_s = 2$ and 3.23 are adopted from Ref. 32. The optimized β values for SKANEX are $\beta = 0.53$ for $r_s = 1$, $\beta = 0.30$ for $r_s = 2$, and $\beta = 0.15$ for $r_s = 3.23$. (d) Electron density distribution along the [110] direction of the cubic simulation cell for the proton configurations shown in (b), obtained from KSDFT simulations and OFDFT calculations using different non-interacting free-energy functionals.

structure factor, and compare the electron density distributions with KSDFT results. In addition, we include results obtained using the standard TF model, the GGA level non-interacting free-energy functional LKTF,²⁸ and the recently developed non-interacting

free-energy functional XWMF,²⁹ which is equivalent to $F_s^{\text{NL}}[n; T]$ in Eq. (4) when $\beta = 1$. These KSDFT and PIMC comparisons are used only as benchmarks for validation; the SKANEX calibration itself uses KSDFT reference electron densities only through the

density-based optimization of β described in Sec. II B. As shown in Fig. 1(c), SKANEX exhibits close agreement with the benchmark KSDFT and PIMC results across all three conditions. In contrast, results obtained using the LKTF and TF functionals show significant discrepancies relative to the KSDFT and PIMC benchmarks. The XWMF data are accurate at $r_s = 1$, but display substantial inaccuracies at $r_s = 2$ and 3.23. Although SKANEX provides substantially better accuracy than the other OFDFT approaches considered, its accuracy deteriorates slightly at $r_s = 3.23$ compared with the higher-density cases. This behavior indicates the onset of the regime in which SKANEX becomes less reliable, as indicated by the blue region in Fig. 1(a), where hydrogen is weakly ionized. However, at such low temperatures, performing standard KSDFT simulations is not problematic, because thermal excitations are relatively weak.

An interesting observation is that the TF model substantially fails to accurately describe the electronic properties at $r_s = 1$ and $T = 50.1$ eV ($\theta = 1$). This corresponds to a fully ionized hot dense plasma regime, for which the TF model is generally considered reliable. The calculations reveal that the TF approximation provides accurate results for the electron-ion static structure factor only in extremely hot plasmas with $T \gtrsim 200$ eV. This is demonstrated in Fig. 2(a), where we compare the SKANEX and TF results for $T = 100.2$ and 200.4 eV at $r_s = 1$ ($\rho = 2.7$ g/cm³). These findings underscore the importance of quantum nonlocality even under such extreme conditions.

As a demonstration of the scalability of SKANEX for electronic properties modeling, Fig. 2(b) shows the electron-ion static structure factor of dense hydrogen at $r_s = 2$ ($\rho = 0.34$ g/cm³) and $\theta = 1$ ($T = 12.5$ eV), computed using systems containing 14, 108, and 1024 ions. The agreement across system sizes demonstrates both the scalability of the method and the robustness of the regularization scheme used for the non-interacting free energy.

In contrast to the electronic properties, the EOS data computed using the TF and LKTF models exhibit comparable quality to SKANEX across the considered temperatures and densities, as summarized in Table I. This table lists pressure values obtained from KSDFT, SKANEX, XWMF, LKTF, and TF calculations. The results clearly demonstrate that even when different functionals give

TABLE I. Pressure (in GPa) of warm dense hydrogen at various densities and temperatures obtained from KSDFT- and OFDFT-based MD simulations. The OFDFT calculations used the TF, XWMF, LKTF, and SKANEX functionals (see Sec. II C for the calculation details). The values of the parameter β used are provided in Table IV (Appendix C).

ρ (g/cm ³)	T (eV)	Pressure (GPa)				
		KS	TF	LKTF	XWMF	SKANEX
2.7	50.1	24 764.4	24 649.6	24 860.0	24 773.2	24 760.5
0.34	12.5	672.8	675.8	678.0	668.7	668.8
0.08	4.8	48.9	52.2	47.7	45.8	45.3

similar pressures, their accuracy for the considered electronic structural properties can differ substantially. In this regard, SKANEX achieves consistently high accuracy for both electron-ion structure factor and EOS calculations. This quality is further illustrated in Fig. 3, which shows the temperature dependence of the electron pressure for a fixed proton lattice, comparing results from KSDFT, SKANEX, and XWMF. SKANEX exhibits excellent agreement with KSDFT over all presented densities and temperatures, except in the low-temperature regime $T \lesssim 4$ eV at $r_s = 3.23$ ($\rho = 0.08$ g/cm³), corresponding to the low-ionization region shown in the lower-left corner of Fig. 1(a).

As an additional demonstration of the utility of SKANEX, in Fig. 4, we re-analyze an X-ray Thomson scattering measurement on warm dense beryllium taken at the NIF by Döppner *et al.*⁴ Specifically, we show the Rayleigh weight $W_R(q)$ quantifying the degree of electronic localization around the nuclei depending on the wavenumber q , and the blue cross corresponds to the experimental measurement obtained in a backscattering geometry with a scattering angle $\alpha = 120^\circ$.³⁵ Independent analyses indicate a temperature of $T \approx 150$ eV,^{4,35,80} making the mass density ρ the main parameter that we would like to infer from the experimental data point. To this end, we have used SKANEX to compute $W_R(q)$ for a broad range of relevant densities; see the different curves in Fig. 4.

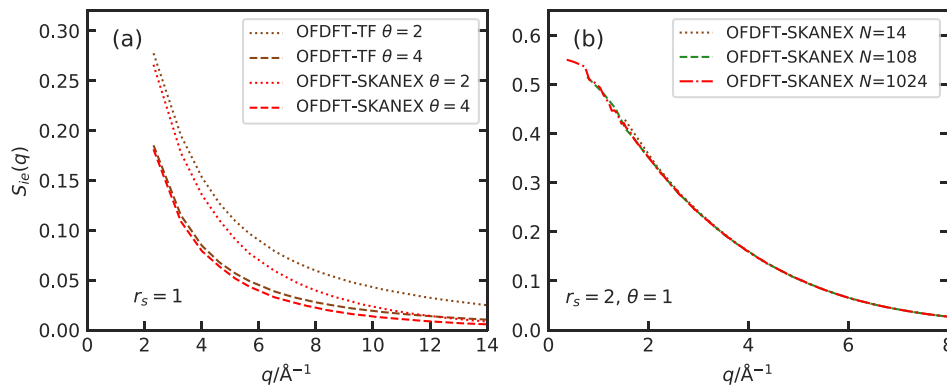


FIG. 2. Electron-ion static structure factor in dense hydrogen plasmas. (a) Comparison of OFDFT results using the TF model and SKANEX at $r_s = 1$ ($\rho = 2.7$ g/cm³) for $\theta = 2$ ($T = 100.2$ eV) and $\theta = 4$ ($T = 200.4$ eV). The optimized β values for SKANEX are 0.59 (for $r_s = 1$ and $\theta = 2$) and 0.68 (for $r_s = 1$ and $\theta = 4$). (b) Comparison of SKANEX result with different numbers of atoms at $r_s = 2$ ($\rho = 0.34$ g/cm³) and $\theta = 1$ ($T = 12.5$ eV), with $\beta = 0.30$.

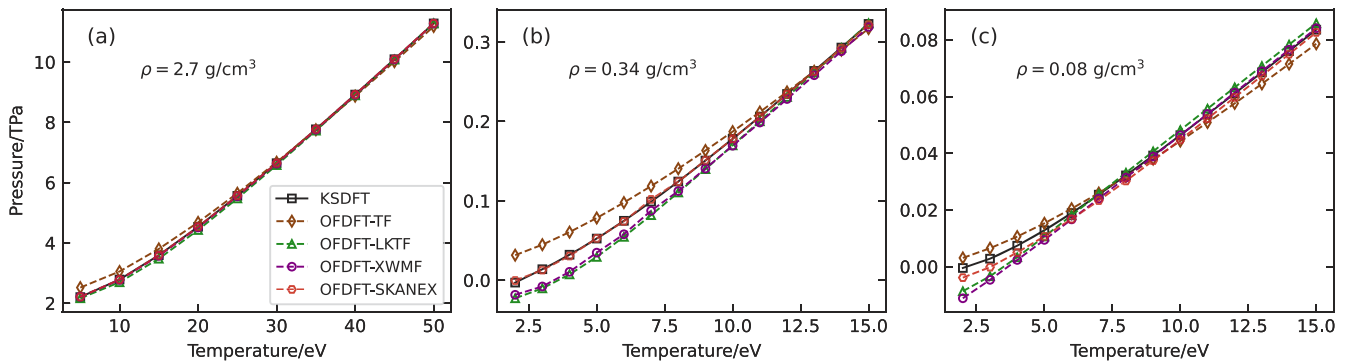


FIG. 3. Electronic pressure for hydrogen atoms arranged in a face-centered cubic lattice, computed using KSDFT and OFDFT at different electron temperatures and at densities of (a) $\rho = 2.7 \text{ g/cm}^3$, (b) $\rho = 0.34 \text{ g/cm}^3$, and (c) $\rho = 0.08 \text{ g/cm}^3$. The values of the parameter β used are provided in Tables VI–VIII (Appendix C).

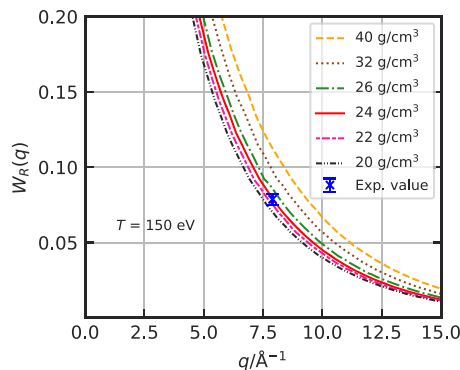


FIG. 4. Rayleigh weight of strongly compressed beryllium at a temperature of 150 eV. OFDFT results and experimental data from NIF measurements are shown.^{4,35} The values of the parameter β used are provided in Table IX (Appendix C).

Our new results indicate a mass density of $\rho = 23 \pm 2 \text{ g/cm}^3$, which is consistent with recent *ab initio* calculations,^{17,35} but substantially lower than indicated by the simpler chemical model used in the original study by Döppner *et al.*⁴ This analysis thus further highlights the great value of SKANEX for practical warm and hot dense matter applications, for which it is capable of delivering state-of-the-art results at a manageable computational cost.

IV. CONCLUSIONS

Advances in high-precision diagnostic tools based on X-ray free-electron lasers, such as those at the European XFEL, have made the development of fast and reliable simulation techniques one of the central challenges in high-energy-density materials research. The OFDFT approach has traditionally been regarded as a suitable method for calculating material properties at high temperatures, on the order of hundreds of thousands of kelvin and hotter. However, beyond reproducing macroscopic observables such as the EOS, a closer examination of different approximations to the

non-interacting free-energy functional reveals that accurate modeling of electronic density and electron–ion structure factors under such extreme conditions remains highly nontrivial. This, in turn, calls for a reassessment of the strategies employed in the development of OFDFT-based methods.

The present work addresses this challenge by introducing a new computational method for the non-interacting free-energy functional, denoted SKANEX, which achieves KSDFT-level accuracy for both the electronic density and the EOS within the orbital-free DFT framework. At the same time, SKANEX retains the substantial computational advantages of the orbital-free approach under extreme conditions, including linear scaling with system size and near-independence of computational cost with temperature. As demonstrations, we have considered warm dense hydrogen and beryllium under experimentally relevant conditions, for which SKANEX is up to several hundred times faster than the standard KSDFT method. For hydrogen, we have demonstrated the importance of quantum nonlocality (i.e., going beyond the local density approximation for the non-interacting free-energy functional) for accurately describing the electron–ion structure factor. For strongly compressed beryllium, using experimental data for the Rayleigh weight, we achieve agreement with much more expensive PIMC calculations regarding the evaluated density range. Extension to heavier elements is also feasible, given recent advances in local pseudopotentials for OFDFT.^{30,81} For regimes dominated by strongly localized bound states, combining the KS-assisted calibration strategy with an $F_s[n]$ better suited for localized states provides a possible route for future improvement. Consequently, SKANEX provides a robust foundation for the development of OFDFT-based simulation tools for electronic density calculations across different regimes. In pursuit of this goal, the presented workflow is planned for inclusion in a forthcoming release of the open-source code DFTpy.²⁴

Uncertainty in the thermodynamic state of high-energy-density experiments generally requires analysis of material properties over a broad range of temperatures and densities.⁸² Performing such systematic studies with KSDFT often requires substantial computational resources, motivating the use of computationally inexpensive but uncontrolled approximations, such as plasma chemical models, which can lead to incorrect interpretations of experimental data.

The considered data for the Rayleigh weight in warm dense beryllium provides a notable example, where originally used chemical models⁴ predicted compressions of $\sim 30 \text{ g/cm}^3$, while both PIMC¹⁷ and independent SKANEX-based OFDFT calculations indicate a substantially lower compression of $\sim 20 \text{ g/cm}^3$. Therefore, a major advantage of the SKANEX-based OFDFT approach is that it enables rapid, systematic exploration of the relevant thermodynamic parameter space, while delivering reliable electronic density information with KSDFT-level accuracy.

Beyond its proven value for the description of equilibrium electronic properties, SKANEX substantially reduces the computational cost of generating statistically converged ionic configurations at extreme temperatures, which are required to calculate transport and optical properties, such as Kubo–Greenwood conductivities and related optical coefficients, as well as for linear-response time-dependent density functional theory. Moreover, the availability of accurate equilibrium electronic properties is a critical prerequisite for extending orbital-free approaches to the time-dependent regime.^{83–85} In this sense, SKANEX provides a reliable foundation for such developments.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Cheng Ma: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Methodology (equal); Software (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Qiang Xu:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Funding acquisition (equal); Methodology (equal); Project administration (equal); Resources (equal); Software (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Zhenhao Zhang:** Data curation (equal); Funding acquisition (equal); Methodology (equal); Project administration (equal); Resources (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Ke Wang:** Data curation (supporting); Methodology (supporting); Validation (supporting); Visualization (supporting). **Ying Sun:** Data curation (supporting); Formal analysis (equal); Funding acquisition (equal); Methodology (equal); Resources (equal); Validation (supporting); Visualization (equal); Writing – review & editing (equal). **Wenhui Mi:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Funding acquisition (equal); Methodology (equal); Project administration (equal); Resources (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Zhandos A. Moldabekov:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Funding acquisition (equal); Methodology (equal); Project administration (equal); Resources (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Tobias Dornheim:** Data curation (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Jan Vorberger:** Data curation (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Resources (equal); Validation (equal); Visualization (equal); Writing – review & editing (equal). **Sebastian Schwalbe:** Data curation (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Resources (equal); Validation (equal); Visualization (equal); Writing – review & editing (equal). **Xuecheng Shao:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Funding acquisition (equal); Methodology (equal); Project administration (lead); Resources (equal); Software (equal); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data supporting the findings of this study are available according to the FAIR principles in the Rossendorf Data Repository (RODARE) at <https://doi.org/10.14278/rodare.4738>, Ref. 86.

TABLE II. Average calculation time per MD step for a 14/32-atom hydrogen system at various densities and temperatures, comparing KSDFT-based MD simulations with MD calculations using SKANEX.

ρ (g/cm ³)	T (eV)	Wall time (s)	
		KSDFT	SKANEX
2.7	50.1	85.75	0.08
0.34	12.5	64.91	0.38
0.08	4.8	30.00	2.39

TABLE III. Single-point energy simulation time for a face-centered cubic cell containing 108 hydrogen atoms at a density of 2.7 g/cm³.

ρ (g/cm ³)	T (eV)	Wall time (s)	
		KSDFT	SKANEX
2.7	5	4.00	0.49
2.7	10	8.53	0.48
2.7	20	25.01	0.48
2.7	30	60.40	0.49
2.7	40	155.31	0.48
2.7	50	328.56	0.47

APPENDIX A: COMPUTATIONAL EFFICIENCY

To evaluate the computational efficiency of SKANEX, we ran two benchmarks on a node equipped with two Intel Xeon Gold 6240R CPUs (24 cores, 2.30 GHz) and 192 GB of RAM. Table II lists the per-step time for a 100-step MD simulation, utilizing only 8 CPU cores for comparison. The computational cost of SKANEX scales linearly with the number of atoms [i.e., $O(N)$] and is only weakly dependent on temperature; at higher temperatures, efficiency may even improve, because there are fewer SCF iterations as a consequence of the system's more homogeneous electron density distribution. By contrast, KSDFT exhibits a much steeper scaling of $O(N^3 T^3)$.⁸⁷ Table III shows a single-point self-consistent field (SCF) calculation for a 108-atom FCC cell using 48 CPU cores. For SKANEX, we excluded the kernel table initialization time (~ 10 s), since it can be read from an input file and is calculated only once at the start of a simulation. As shown in the data, OFDFT is ~ 10 –1000 times faster per SCF step than KSDFT methods, and the OFDFT timing is also temperature-independent.

APPENDIX B: COMPUTATIONAL COST OF β OPTIMIZATION

The optimization of β involves three main steps. First, an XWMF-MD simulation of ~ 10 000 steps is performed to sample representative ionic configurations. The computational cost per MD step is comparable to the SKANEX timings reported in Tables II and III. Second, KSDFT single-point calculations are performed for 10 randomly selected uncorrelated ionic configurations, with the cost per configuration comparable to the KSDFT timings reported in Tables II and III. Third, the parameter β is optimized using

TABLE IV. Optimized β values for warm dense hydrogen under the main MD benchmark conditions discussed in Fig. 1.

ρ (g/cm ³)	T (eV)	β	Standard deviation
2.7	50.1	0.53	0.026
0.34	12.5	0.30	0.018
0.08	4.80	0.15	0.005

TABLE V. Optimized β values for warm dense hydrogen conditions corresponding to the results presented in Fig. 2(a).

ρ (g/cm ³)	T (eV)	β	Standard deviation
2.7	100.2	0.59	0.033
2.7	200.4	0.68	0.030

TABLE VI. Optimized β values for hydrogen at $\rho = 2.7$ g/cm³ used in the temperature-dependent EOS calculations corresponding to the results presented in Fig. 3(a).

ρ (g/cm ³)	T (eV)	β	Standard deviation
2.7	5	0.65	0.034
2.7	10	0.55	0.021
2.7	15	0.50	0.025
2.7	20	0.48	0.028
2.7	25	0.48	0.029
2.7	30	0.49	0.029
2.7	35	0.50	0.028
2.7	40	0.51	0.028
2.7	45	0.52	0.026
2.7	50	0.53	0.026

TABLE VII. Optimized β values for hydrogen at $\rho = 0.34$ g/cm³ used in the temperature-dependent EOS calculations corresponding to the results presented in Fig. 3(b).

ρ (g/cm ³)	T (eV)	β	Standard deviation
0.34	2	0.40	0.030
0.34	3	0.32	0.025
0.34	4	0.29	0.021
0.34	5	0.28	0.019
0.34	6	0.27	0.018
0.34	7	0.28	0.018
0.34	8	0.28	0.018
0.34	9	0.28	0.018
0.34	10	0.29	0.018
0.34	11	0.30	0.018
0.34	12	0.30	0.018
0.34	13	0.31	0.018
0.34	14	0.31	0.018
0.34	15	0.32	0.018

TABLE VIII. Optimized β values for hydrogen at $\rho = 0.08 \text{ g/cm}^3$ used in the temperature-dependent EOS calculations corresponding to the results presented in Fig. 3(c).

$\rho \text{ (g/cm}^3\text{)}$	$T \text{ (eV)}$	β	Standard deviation
0.08	2	0.14	0.005
0.08	3	0.14	0.004
0.08	4	0.15	0.005
0.08	5	0.15	0.005
0.08	6	0.16	0.006
0.08	7	0.17	0.007
0.08	8	0.18	0.007
0.08	9	0.19	0.008
0.08	10	0.20	0.008
0.08	11	0.21	0.008
0.08	12	0.21	0.009
0.08	13	0.22	0.009
0.08	14	0.23	0.010
0.08	15	0.24	0.010

TABLE IX. Optimized β values for beryllium at $T = 150 \text{ eV}$ corresponding to the results presented in Fig. 4.

$\rho \text{ (g/cm}^3\text{)}$	$T \text{ (eV)}$	β	Standard deviation
20	150	0.19	0.004
22	150	0.20	0.004
24	150	0.21	0.005
26	150	0.21	0.004
32	150	0.23	0.006
40	150	0.25	0.003

the Powell algorithm, which requires dozens of OFDFT single-point calculations for each temperature–density combination. As an example, the optimization of β for warm dense hydrogen at $\rho = 2.7 \text{ g/cm}^3$ and $T = 50.1 \text{ eV}$ required ~ 50 min on a single processor for this final step. Overall, the full β optimization procedure requires only a few hours of computation, which is substantially less expensive than a full KSDFT–MD production run, which typically requires several days.

APPENDIX C: USED β PARAMETERS

For each material and thermodynamic condition, the SKANEX parameter β was optimized by minimizing the integrated absolute deviation between the OFDFT and KSDFT electron densities. To account for configurational variation, $N_i = 10$ ionic configurations were sampled from an OFDFT–MD trajectory generated with XWMF for each condition. For each configuration, a separate optimal parameter, β_i , was obtained. The β value reported in Tables IV–IX corresponds to the arithmetic mean of these β_i values, while the associated uncertainty is given by their standard deviation. For face-centered cubic lattice EOS calculations, the β parameters were optimized using the same ionic configuration as in the MD simulations at $\theta = 1$.

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